

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	15	16	17	18	19	20	21	22	23	24	25	26	27	28
U	29	30	31											
G	M	T	W	T	F	S	S	M	T	W	T	F	S	S

Theory - 4

FRIDAY

JULY

15

WK 29 • 196-169

Cipher text: H L D T Q I H J D W X Q C M A G, Decryptkey = $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$ find plain text (A=0, B=1)→ determinant $|k| = (3 \times 7) - (2 \times 5)$ inverse of determinant $|k|^{-1} \text{ mod } 26$ $[\det k^{-1}] \text{ adj} \begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix}$

$$26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$1 = 4 - 3$$

$$1 = 4 - (11 - 4 \cdot 2)$$

$$1 = 4 - 11 + 4(2)$$

$$1 = 3(4) - 11$$

$$1 = 3(26 - 11(2)) + 11$$

$$1 = 3(26) - 11(6) + 11$$

$$19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \text{ mod } 26$$

$$\begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \text{ mod } 26$$

$$\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} //$$

$$\begin{array}{c|c|c|c|c|c|c}
 H & L & D & T & Q & I & H & J & D & W & X & Q & C & M & A & G \\
 7 & 11 & 3 & 19 & & & & & & & & & & & &
 \end{array}$$

$$= 26 - 7$$

$$= 19$$

$$\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3(7) + (14 \times 11) \\ (9 \times 7) + (5 \times 11) \end{bmatrix} = \begin{bmatrix} 19 \\ 14 \end{bmatrix} \begin{matrix} T \\ O \end{matrix}$$

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SATURDAY

JULY

197-168 • WK 29

1	2	3	4	5	6	7	8	9	10	11	12
13	14	15	16	17	18	19	20	21	22	23	24
25	26	27	28	29	30						
M	T	W	T	F	S	S	M	T	W	T	F
S	S										

J
U
N
'22

$$\begin{aligned}
 \text{iii)} \quad \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \pmod{26} &= \begin{bmatrix} 3(3) + 14(19) \\ 9(3) + 5(19) \end{bmatrix} \pmod{26} \\
 &= \begin{bmatrix} 275 \\ 122 \end{bmatrix} \pmod{26}
 \end{aligned}$$

$$\begin{bmatrix} 15 \\ 18 \end{bmatrix} = \begin{matrix} P \\ S \end{matrix}$$

$$\begin{aligned}
 \text{iii)} \quad \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 22 \end{bmatrix} \pmod{26} &= \begin{bmatrix} 3 \times 16 + 14 \times 22 \\ 9 \times 16 + 5 \times 22 \end{bmatrix} \pmod{26} = \begin{bmatrix} 160 \\ 184 \end{bmatrix} \pmod{26} \\
 &= \begin{bmatrix} 4 \\ 2 \end{bmatrix} \begin{matrix} E \\ C \end{matrix}
 \end{aligned}$$

$$\begin{aligned}
 \text{iv)} \quad \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} \pmod{26} &= \begin{bmatrix} 3(7) + 14(9) \\ 9(7) + 5(9) \end{bmatrix} \pmod{26} \\
 &= \begin{bmatrix} 147 \\ 108 \end{bmatrix} \pmod{26} = \begin{bmatrix} 17 \\ 4 \end{bmatrix} \begin{matrix} R \\ E \end{matrix}
 \end{aligned}$$

$$\begin{aligned}
 \text{v)} \quad \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} &= \begin{bmatrix} 3(3) + 14(23) \\ 9(3) + 5(23) \end{bmatrix} \pmod{26} \\
 &= \begin{bmatrix} 331 \\ 142 \end{bmatrix} \pmod{26} = \begin{bmatrix} 9 \\ 12 \end{bmatrix} \begin{matrix} T \\ M \end{matrix}
 \end{aligned}$$

$$\begin{aligned}
 \text{vi)} \quad \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} &= \begin{bmatrix} 3(22) + 14(16) \\ 9(22) + 5(16) \end{bmatrix} \pmod{26} \\
 &= \begin{bmatrix} 290 \\ 278 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 18 \end{bmatrix} \begin{matrix} E \\ S \end{matrix}
 \end{aligned}$$

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$$\begin{aligned}
 \text{vi)} \quad \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} &= \begin{bmatrix} 3(22) + 14(16) \\ 9(22) + 5(16) \end{bmatrix} \pmod{26} \\
 &= \begin{bmatrix} 290 \\ 278 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 18 \end{bmatrix} \begin{matrix} E \\ S \end{matrix}
 \end{aligned}$$

2022

$$\text{vii)} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} = \begin{bmatrix} 3(2) + 14(12) \\ 9(2) + 5(12) \end{bmatrix} = \begin{bmatrix} 174 \\ 78 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 18 \\ 0 \end{bmatrix} \begin{matrix} S \\ A \end{matrix}$$

$$\text{viii)} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 3(0) + 14(6) \\ 9(0) + 5(6) \end{bmatrix} = \begin{bmatrix} 84 \\ 30 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 6 \\ 4 \end{bmatrix} \begin{matrix} G \\ E \end{matrix}$$

Plaintext: TOPSECRETMESSAGE

2) Consider the Hill Cipher key Matrix

$$k = \begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix} \text{ Cipher text : RXADPY Find plaintext}$$

$$\rightarrow \begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix} = (24 \times 14) - (5 \times 19) = \frac{241}{26} = 9.269$$

$$= 7$$

$$7^{-1}$$

$$26 = 7(3) + 5, \quad 5 = 26 - 7(3)$$

$$7 = 5(1) + 2, \quad 2 = 7 - 5(1)$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2)$$

$$1 = 5 - 2(7 - 5(1))$$

$$1 = 3(5) - 2(7)$$

$$1 = 3(26 - 7(3)) - 2(7)$$

$$15 \begin{bmatrix} 14 & 14 \\ -5 & 24 \end{bmatrix} = \begin{bmatrix} 210 & 285 \\ -75 & 370 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

19

TUESDAY

JULY

200-165 • WK 30

1	2	3	4	5	6	7	8	9	10	11	12
13	14	15	16	17	18	19	20	21	22	23	24
25	26	27	28	29	30						
M	T	W	T	F	S	S	M	T	W	T	F

$$1^2 = (-11)7 + 3(26)$$

$$-11 + 26 = 15 //$$

$$i) \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 2(17) + 1(23) \\ 3(17) + 22(23) \end{bmatrix} = \begin{bmatrix} 547 \\ 557 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 5 \\ 11 \end{bmatrix} = \begin{matrix} F \\ L \end{matrix}$$

$$ii) \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 2(0) + 1(14) \\ 3(0) + 22(14) \end{bmatrix} = \begin{bmatrix} 14 \\ 308 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{matrix} O \\ W \end{matrix}$$

$$iii) \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 2(3) + 1(24) \\ 3(3) + 22(24) \end{bmatrix} = \begin{bmatrix} 30 \\ 537 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 4 \\ 17 \end{bmatrix} = \begin{matrix} E \\ R \end{matrix}$$

Plain text : FLOWER

6 Relation between substitution, confusion, ^{transposition} ~~transposition~~ & Diffusion

→ Substitution & Confusion :- It replaces one character with another character.

Ex :- HELLO → KHOOR

The relationship is :-

2022

Substitution creates confusion by hiding

the relationship between original plaintext character & ciphertext character

so, Substitution $\rightarrow$ Confusion, makes it difficult to understand how the key & plaintext are related to the ciphertext.

Transposition $\rightarrow$ Diffusion, Transposition changes the position of character without changing the character themselves

Ex:- HELLO $\rightarrow$ LHDEL

The relationship is:

Transposition achieves diffusion by spreading the influence of plaintext characters across different positions in ciphertext.

So, Transposition $\rightarrow$ Diffusion

Diffusion spreads influence of plaintext throughout the ciphertext making pattern harder to recognize

2) Analyze classical cipher with respect to confusion & diffusion. Complete the table & justify your classification based on how each cipher transform plaintext

Cipher	Does it change character/symmetric value	change position	confusion/ Diffusion
Shift Cipher	Yes	No	2022 Confusion

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THURSDAY

JULY

202-163 • WK 30

1 2 3 4 5 6 7 8 9 10 11 12
 13 14 15 16 17 18 19 20 21 22 23 24 25 26
 27 28 29 30
 M T W T F S S M T W T F S S

8 Justification: Each letter is shifted by fixed by fixed key

9 Ex: ABC with key 3. The value change, but pos, how DEF remain same. Then hide original sym-to-sym relationship

11	Multiplicative Cipher	Yes	No	Confusion
12				

1 Justification: Each letter value is multiplied by key & taken mod 26.
 2 Ex: Key 5, $A(0) \rightarrow A(0)$
 3 $B(1) \rightarrow B(5)$. The character value changes, while same position.

4	Affine Cipher	Yes	No	Confusion
5				

6 Justification: Each character transformed using $C = (aP + b) \bmod 26$ both multiplication & addition change the character
 Value $A(0) \rightarrow D(3)$. When $a \neq 1$, $b \neq 3$

Transposition	No	Yes	Diffusion
---------------	----	-----	-----------

The character themselves does not change, only their positions are re-arranged.
 Ex: HELLO → LHOEL This spreads the original character segments & makes plaintext pattern harder to recognize.

Transposition Cipher (Classical):

It is a cryptographic technique in which positions of characters are re-arranged according to a specific rule or key.

- The character themselves does not change
- Only their order/position changes
- It is mainly associated with diffusion

Ex: HELLO → LHOEL

Syntax: $C = T_k(P)$

the letters are same but position changes

Decryption: $P = T_k^{-1}(C)$

Where: P = plaintext

C = cipher

k = key

T_k = Transposition operation using key k .

23

SATURDAY

JULY

204-161 • WK 30

1	2	3	4	5	6	7	8	9	10	11	12
13	14	15	16	17	18	19	20	21	22	23	24
25	26	27	28	29	30						
M	T	W	T	F	S	S	M	T	W	T	F
S	S										

Rules:- do not change character values /
change only position / order

Steps:- Remove space.

• Write key / keyword at top

• Number the letter according to alphabetical order

• Write plaintext under key
in row - plain text
head column - cipher

Combine Column operation according to key order

Pie lnc att

Transposition

a) Encryption: The arrow is pointing north direction using key

24 SUNDAY

Remove space:-

THE ARROW IS POINTING NORTH DIRECTION
key = 4 - 4 column

2022

Column

	1	2	3	4
8	1 : TRIINRI	T	H	E
	2 : HRSNGTRI	R	R	O
9	3 : EOPTNHEO	I	S	P
	4 : AWOIODCN	I	N	T
10		N	G	N
		R	T	H
11		I	R	E
		T	I	O
12				N

Cipher text:-

1 TRIINRI THRSNG TRIE OPTNHEO
 AWOIODCN

2 b) Keyword :- ATLAS

3 The arrow is pointing north direction

4 Removes space

keyword:- A T L A S
 1 5 3 2 4

5 Column:-

1 : TRPFRRO	T	H	E	A	R
2 : AINNPT	R	O	W	I	S
3 : EWIGHC	P	O	I	N	T
4 : RSTOIT	I	N	G	N	O
5 : HOONTEN	R	T	H	D	I
	R	E	C	T	I
	O	N	X	X	X

26

TUESDAY

JULY

207-158 • WK 31

1 2 3 4 5 6 7 8 9 10 11 12
13 14 15 16 17 18 19 20 21 22 23 24 25 26
27 28 29 30
M T W T F S S M T W T F S S

double key 1: BREAK
key 2: LOG

key: B R E A K
2 S 3 1 4

1 → A: A I N N D T

2 → B: T R P I R R O

3 → E: E W I G H C

4 → K: R S T O I

5 → R: H O O N T E N

T H E A R

R O W I S

P O I N T

T N G N O

R T H D T

R E C T I

O N X X X

1st Transposition

A I N N D T T R P I R R O

E W I G H C R S T O I

H O O N T E N

key 2: L O G
2 3 1

A I N

N D T

T R P

I R R

O B W

T G H

C R S

T O I

N T

1: G: N T R R W H S T O B

2: L: A N T I O R C T H N

3: O: I T D R R E G R O O T

N T P R W S I O E A N T I O R C

T H N N I D R R E G R O O T

Decryption:-

1) Ciphertext: ~~KIDT~~ TIDIE MRNME REOIO
 SANOW RSLTP EEEDO STTNV
 AHTUD BLEND GODAB
 PETXD ELNSX

key=10

$60 \div 10 = 6$ rows - fill column wise

	1	2	3	4	5	6	7	8	9	10
1	T	R	O	O	P	S	H	E	A	D
2	I	N	G	W	E	S	T	N	E	E
3	D	M	O	R	E	S	U	P	P	L
4	I	E	S	S	E	M	D	G	E	N
5	E	R	A	L	D	U	B	O	I	S
6	M	E	N	T	O	A	I	D	X	X

Step:- Read Row-wise.

TROOPS HEADING WEST NEED MORE
 SUPPLIES SEND GENERAL DUBOIS
 MEN TO AID

2) Cipher: ILEDT NHDTR SDIIT MAXSA
 XERYT ANONT SNFOT X

keyword:- SECRET

521436

36 letter

$36 \div 6 = 6$ rows

28

THURSDAY

JULY

209-156 • WK 21

1 2 3 4 5 6 7 8 9 10 11 12
 13 14 15 16 17 18 19 20 21 22 23 24 25 26
 27 28 29 30
 M T W T F S S M T W T F S S

First

	S	B	C	R	E	T
8	5	2	1	4	3	6
1	T	H	I	S	I	S
2	A	P	L	A	I	N
3	N	T	E	X	T	F
4	O	R	D	E	M	O
5	N	S	T	R	A	T
6	I	O	N	X	X	X

Fill column wise in order

1 → 2 → 3 → 4 → 5 → 6

Read-row wise THIS APLAIN NTXEXT
 ORDEMO NSTRTAT
 IONXXX
 ⇒ THIS IS A PLAIN NTEXT FOR
 DEMONSTRATION

3) Ciphertext:- VEY FAXARA ATX HGD

keyword:- Plain

Total letter = $15 \div 3 = 3$ rows

	P	L	A	I	N
	5	3	1	2	4
1	H	A	V	E	A
2	G	R	E	A	T
3	D	A	Y	X	X

read row-wise

HAVE A GREAT DAY

∴ HAVE A GREAT DAY

22

WK 31 210-155

Find Inverse permutation of $P(4, 3, 1, 5, 2)$

Position	1	2	3	4	5	Where is	in position
P	4	3	1	5	2	1	3
	3	5	2	1	4	2	5
						3	2
						4	1
						5	4

$$P^{-1} = (3, 5, 2, 1, 4)$$

2) $P = (7, 4, 1, 3, 5, 2, 6)$

POS	1	2	3	4	5	6	7	Where is	in position
P	7	4	1	3	5	2	6	1	3
P^{-1}	3	6	4	2	5	7	1	2	6

$$P^{-1} = (3, 6, 4, 2, 5, 7, 1)$$

3) $P = (4, 1, 3, 2)$

POS	1	2	3	4	Where is	in position
P	4	1	3	2	1	2
P^{-1}	2	4	3	1	2	4

$$P^{-1} = (2, 4, 3, 1)$$

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SATURDAY

JULY

211-154 • WK 31

1	2	3	4	5	6	7	8	9	10	11	12
13	14	15	16	17	18	19	20	21	22	23	24
25	26	27	28	29	30						
M	T	W	T	F	S	S	M	T	W	T	F
S	S										

u) $P_2(3, 5, 1, 4, 2)$

8

pos 1 2 3 4 5

9

P 3 5 1 4 2

$P^{-1}(3, 5, 1, 4, 2)$

10

P^{-1} 3 5 1 4 2

11

12

1

2

3

4

5

6

31 SUNDAY